

MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY
(An Autonomous Institution)

DETAILS OF THE PROJECT

Title of the Project	Intelligent Vehicle Speed Limiting and Monitoring System Using Hybrid IR-LiFi-LoRa Communication
Batch Number	A-14
Department	Information Technology
Team Members	DHARSHNAMOORTHY R REG.NO: 22TH0223 GOGULAKANNAN M REG.NO: 22TH0228 HENRY DANIEL DIDACE D REG.NO: 22TH0232 PRASHANTH ARJUN GK REG.NO: 22TH0276
Year / Duration	2025-26 / 6 months
Problem Statement	Road accidents caused by over-speeding remain a critical safety concern, particularly in sensitive zones such as school areas, hospitals, residential streets, and construction sites. Existing speed regulation mechanisms - including speed breakers, static signboards, radar-based surveillance, and GPS-assisted advisory systems — are either static, reactive, or dependent on expensive and continuous network infrastructure. They lack the ability to autonomously enforce speed limits in real time without driver intervention. Furthermore, advanced V2X and cellular-based systems are cost-prohibitive for large-scale deployment in developing regions. There is a clear need for a low-cost, autonomous, and reliable system that can dynamically enforce vehicle speed limits in restricted zones without relying on continuous high-bandwidth connectivity or driver compliance.
Objectives of the Project	This project presents an intelligent vehicular speed control system that leverages a hybrid communication approach combining Infrared (IR), Light Fidelity (LiFi), and Long Range (LoRa) technologies. The system consists of a roadside unit, which detects vehicle presence via an IR sensor and transmits the permissible speed

	limit using LiFi-based optical communication, while the vehicle unit receives, processes, and enforces the limit through PWM-based motor control, and LoRa is used for telemetry purposes. • To design and implement a hybrid IR-LiFi-LoRa communication-based system that automatically enforces vehicle speed limits in restricted zones without driver intervention. • To utilize LiFi-based optical communication for real-time, low-cost speed limit transmission from roadside units to vehicles. • To implement PWM-based motor speed control for smooth and automatic speed regulation upon entering restricted zones. • To enable remote monitoring of vehicle speed and system status via LoRa communication to a monitoring station. • To provide a fail-safe mechanism and manual override feature for safety under signal loss or emergency conditions.
Tools & Platforms	Hardware: • ESP32 Microcontrollers • IR Sensor • High-brightness LED (LiFi Transmitter) • Phototransistor (LiFi Receiver) • L298N Motor Driver • DC Motors (RC Car) • LoRa Module (SX1278) • Push Button (Override) • Buzzer • Battery Pack Software: • Visual Studio Code with PlatformIO, Embedded C / Arduino Framework, LoRa Library (SX1278), PWM / LEDC Functions (ESP32) • Cloud IoT Platform • Mobile Dashboard

PO's / PSO's Mapped	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12 / PSO2
SDGs Mapped	• SDG 3 – Good Health and Well-Being • SDG 9 – Industry, Innovation and Infrastructure • SDG 11 – Sustainable Cities and Communities
SDG Justification (in 100 words)	SDG 3: Good Health and Well-Being: This project directly contributes to SDG 3 by reducing road accidents caused by over-speeding in sensitive zones such as schools and hospitals, thereby protecting human lives. SDG 9: Industry, Innovation and Infrastructure: It aligns with SDG 9 through the development of an innovative, low-cost hybrid communication system combining IR, LiFi, and LoRa technologies for intelligent transportation. SDG 11: Sustainable Cities and Communities: It supports SDG 11 by enabling smarter, safer urban mobility without reliance on expensive infrastructure, making it scalable for smart city environments and developing regions.

HEAD OF THE DEPARTMENT
(Dr. P. SIVAKUMAR)

PROJECT GUIDE
(Mrs. V. VIMALA DHEEKSHANYA)